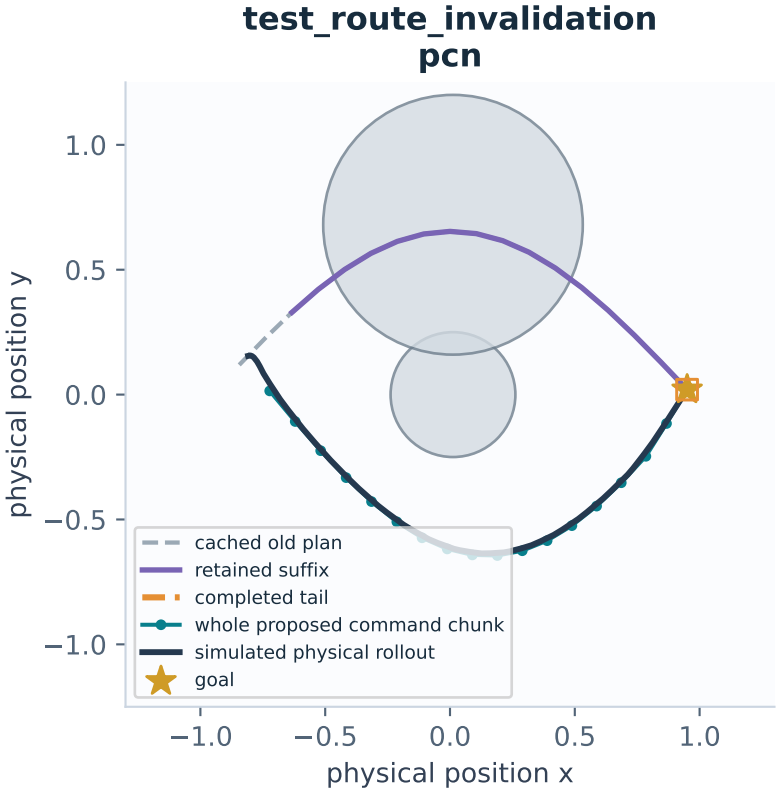MCMC replaces a complete future chunk; physical execution follows its selected prefix